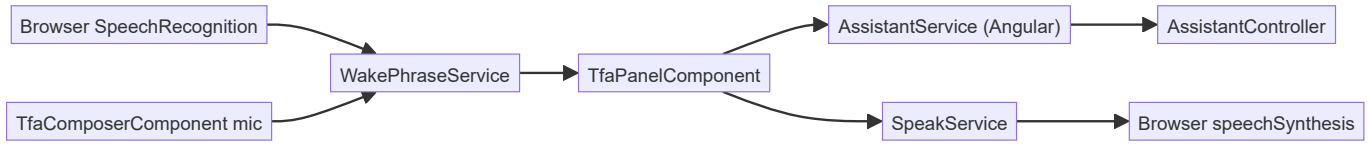
Voice Input – Technical Documentation

> **Project:** TaskFlow – Intelligent Task Allocation & Tracking System **Parent Module:** TaskFlow Assistant (TFA) · **Child Module:** 09.5 Voice Input **Version:** 1.0 · **Last Updated:** 31-Oct-2026 **Backend:** ASP.NET Core Web API (.NET 10) · **Frontend:** Angular · **Database:** SQLite (EF Core) **Author:** Jojo George (jojogeorge3344@gmail.com) © 2026 Jojo George. All rights reserved.

1. Technical Overview

Voice Input (hands-free) is frontend only. `WakePhraseService` wraps the browser `SpeechRecognition` / `webkitSpeechRecognition` API to listen for the wake phrase "Hey TF assistant" and for dictated commands; `SpeakService` wraps `speechSynthesis` to read answers aloud. `TfaPanelComponent` wires both to the panel, and `TfaComposerComponent` provides the mic button and hands-free toggle. Recognised text is sent through the normal chat endpoint (09.1); the backend cannot tell a spoken question from a typed one.

2. Architecture Flow



3. Backend File Structure

Layer	Path	Files
–	–	*Not applicable – no voice-specific backend files; text goes through `AssistantController` / `AssistantService` (09.1).*

4. Frontend File Structure

Type	Path	Purpose
Voice services	`src/app/modules/assistant/services/wake-phrase.service.ts`	Browser `SpeechRecognition` (wake phrase, dictation; `localStorage` key `tfa.handsfree`)
Voice services	`src/app/modules/assistant/services/speak.service.ts`	`speechSynthesis` (max 500 chars spoken)
Models	`src/app/modules/assistant/services/wake-phrase.model.ts`, `services/speak.model.ts`	TypeScript interfaces for the speech factories
Panel component	`src/app/modules/assistant/components/tfa-panel.component.ts/.html`	Voice wiring, `answerAloud`, voice hint and heard text under the TFA button
Child components	`src/app/modules/assistant/components/tfa-composer.component.ts/.html`	Mic button, listening bar, hands-free toggle

5. API & Logic

5.1 API Endpoints

#	Method	Endpoint	Description	Permission (Policy)	Request DTO	Response DTO
–	–	(frontend only)	Recognised text is sent through `POST /api/v1/assistant/conversation/messages`	`perm:assistant.chat`	`SendAssistantMessageRequest`	`AssistantReplyDto`

5.2 Request / Response Samples

```
// Spoken "Hey TF assistant, what are my upcoming tasks" becomes
{ "text": "what are my upcoming tasks", "provider": "Ollama" }
```

5.3 Service Layer Logic

`WakePhraseService` (`wake-phrase.service.ts`)

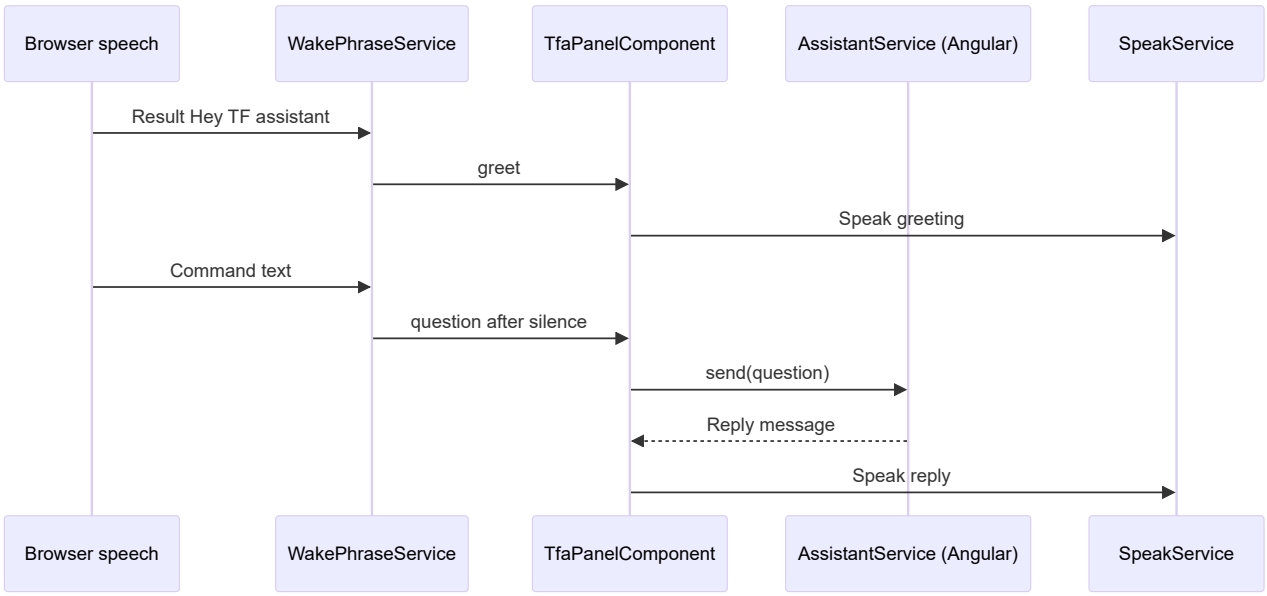
1. `SPEECH_RECOGNITION_FACTORY` creates `SpeechRecognition` / `webkitSpeechRecognition` or null (unsupported).
2. Hands-free listens continuously for `WAKE_PHRASE` (regex tolerant of "Hey T F assistant", "Hay TFA assistant", "Hey Tifa assistant").
3. Emits `wake`, `greet` (phrase alone), `interim` (live text) and `question` (after `COMMAND_SILENCE_MS` 2000 ms of silence). Capture timeout `WAKE_CAPTURE_TIMEOUT_MS` 8000 ms; mic-button dictation timeout `DICTATION_TIMEOUT_MS` 10000 ms.
4. Restarts ended sessions after 50 ms; quick failures back off from 300 ms and give up after 6 with a reason message.
5. Only a deliberate turn-off is persisted (`tfa.handsfree`).

`SpeakService` (`speak.service.ts`): `speechSynthesis` wrapper (`SPEECH_SYNTHESIS_FACTORY`, `UTTERANCE_FACTORY`); max `MAX_SPOKEN_CHARS` 500; `TfaPanelComponent.answerAloud` speaks only replies to spoken questions and pauses recognition while speaking.

5.4 Repository Layer

Method	Description	Query Notes
–	No backend storage	Browser `localStorage` only

5.5 Sequence Diagram



6. Database Design

Not applicable – no tables; only the browser `localStorage` key `tfa.handsfree` is stored.

7. Security & Authorization

- Voice controls are part of the TFA Panel, rendered only for holders of `assistant.chat` (shell).
- Speech-to-text is done by the browser's own speech service; TaskFlow keeps no audio. Spoken text passes through the same server-side checks as typed text (length, `SecretScreen`, rate limit).

Permission Key	Endpoint(s)	UI Element Controlled
`assistant.chat`	`POST /api/v1/assistant/conversation/messages`	Mic button, listening bar, hands-free toggle, wake phrase

8. Validations & Error Handling

Rule	Layer	HTTP Code	Error Message
Browser has no speech recognition	`WakePhraseService` (factory returns null)	–	Voice input is not supported in this browser. Try Chrome or Edge, or type your question.
Error `not-allowed` / `service-not-allowed` (no auto start)	`WakePhraseService.onError`	–	Microphone access was denied. Allow it in your browser to use voice, or type your question.
Error `audio-capture`	`WakePhraseService.onError`	–	No microphone was found. You can type your question instead.
Error `network`	`WakePhraseService.onError`	–	Voice needs an internet connection in this browser. You can type your question instead.
Error `not-allowed` while auto-starting from an earlier visit	`WakePhraseService.waitForGesture`	–	Click anywhere on the page to turn "Hey TF assistant" back on.

Rule	Layer	HTTP Code	Error Message
`no-speech` / `aborted`	`WakePhraseService.onError`	–	Ignored (normal while waiting)
Repeated quick session ends	`WakePhraseService.onEnd` / `retryLater`	–	Gives up after 6 attempts with a reason message (not persisted as "off")

9. Integrations / Side Effects

- **Browser APIs:** Web Speech API (`SpeechRecognition` , `speechSynthesis`).
- **Notifications / Email / Activity Log:** none.

10. Configuration

Not applicable – no `appsettings.json` keys. Frontend constants: `COMMAND_SILENCE_MS` 2000, `WAKE_CAPTURE_TIMEOUT_MS` 8000, `DICTATION_TIMEOUT_MS` 10000, `MAX_SPOKEN_CHARS` 500; `localStorage` key `tfa.handsfree` (`TFA_VOICE_KEY`).

11. Testing

Test Type	File / Location	Covers
Frontend unit	`src/app/modules/assistant/services/wake-phrase.service.spec.ts`	Voice is on by default and remembers when the user turns it off; pause rests the microphone without turning voice off; "Hey TF assistant" (common spellings and mishearings, unfinished or in one breath) opens TFA and hands over the question without sending it; the mic button takes one command and closes, unless hands-free is on; long commands in pieces are joined and sent once; it returns to standby and times out waiting; one session at a time with a fresh recognizer, back-off and a named reason when sessions keep stopping; unsupported browser, denied or missing microphone and network errors are explained, silence is normal
Frontend unit	`src/app/modules/assistant/services/speak.service.spec.ts`	Reads text aloud and calls done once, also when the engine never reports the end (safety limit) or has no speech engine; says nothing for empty text; cancel stops speech; `toSpeech` drops markdown and bare addresses, reads links by label and cuts long replies at whole sentences
Frontend unit	`src/app/modules/assistant/components/tfa-panel.component.spec.ts`	Voice block: listens for "Hey TF assistant" from the start with no animation until a command is taken; opens on speech, sends the question and returns to standby; answers a second wake phrase; does not use the microphone when voice is off and listens again when turned on; the mic button command flow; voice state and heard text on the TFA button; spoken questions get a text and spoken answer with the microphone rested, typed ones are not read aloud, failures are spoken; unsupported browser and denied microphone are explained and typing still works; a blocked microphone after reload asks for a click

12. Known Limitations / Gaps found

- No voice-specific gaps were recorded in the existing review. Voice depends on browser support (Chrome, Edge).

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